

0.4 Equivalence Relations

In mathematics, we often want to say that two objects are “the same” in some sense, even if they are not identical. For instance:

- Two fractions $\frac{1}{2}$ and $\frac{2}{4}$ represent the same rational number.
- Two matrices **A** and **B** are **similar** if $\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ for some invertible **P**—they represent the same linear transformation in different bases.
- Two vector spaces are **isomorphic** if there exists a bijective linear map between them—they have the same “structure.”

Each of these notions of “sameness” is captured by an **equivalence relation**. Understanding equivalence relations helps us recognize when different-looking objects are fundamentally alike.

0.4.1 Relations

Definition 0.4.1 — Relation

A **(binary) relation** on a set A is a subset $R \subseteq A \times A$. If $(a, b) \in R$, we write $a \sim b$ (or $a R b$) and say “ a is related to b .”

Example. (Examples of Relations)

Here are some common examples of relations:

1. On $\mathbb{Z}$, define $a \sim b$ if $a \leq b$. This is the “less than or equal to” relation.
2. On $\mathbb{R}$, define $a \sim b$ if $|a - b| < 1$. This says a and b are “close.”
3. On the set of all people, define $a \sim b$ if a and b have the same birthday.

0.4.2 Equivalence Relations

Not all relations behave like “sameness.” An **equivalence relation** is one that satisfies three natural properties that any notion of sameness should have.

Definition 0.4.2 — Equivalence Relation

A relation $\sim$ on a set A is an **equivalence relation** if it satisfies:

1. **Reflexivity:** $\forall a \in A : a \sim a$.
(Every element is related to itself.)
2. **Symmetry:** $\forall a, b \in A : a \sim b \Rightarrow b \sim a$.
(If a is related to b , then b is related to a .)
3. **Transitivity:** $\forall a, b, c \in A : (a \sim b \wedge b \sim c) \Rightarrow a \sim c$.
(If a is related to b and b is related to c , then a is related to c .)

Example. (Examples of Equivalence Relations)

The following are all equivalence relations:

1. **Equality:** On any set A , the relation $a \sim b \iff a = b$ is an equivalence relation (the finest one possible).
2. **Congruence modulo n :** On $\mathbb{Z}$, define $a \equiv b \pmod{n}$ if $n \mid (a - b)$.
 - Reflexive: $n \mid 0$, so $a \equiv a$.
 - Symmetric: If $n \mid (a - b)$, then $n \mid (b - a)$.
 - Transitive: If $n \mid (a - b)$ and $n \mid (b - c)$, then $n \mid ((a - b) + (b - c)) = (a - c)$.
3. **Same birthday:** On the set of people, “ a has the same birthday as b ” is an equivalence relation.
4. **Same absolute value:** On $\mathbb{R}$, define $a \sim b \iff |a| = |b|$. This is an equivalence relation.

Example. (Non-Examples)

The following relations fail to be equivalence relations:

1. On $\mathbb{Z}$, the relation $a \leq b$ is **not** an equivalence relation:
 - Reflexive: Yes, $a \leq a$.
 - Symmetric: **No.** $1 \leq 2$ but $2 \not\leq 1$.
2. On $\mathbb{R}$, the relation $|a - b| < 1$ is **not** an equivalence relation:
 - Reflexive: Yes, $|a - a| = 0 < 1$.
 - Symmetric: Yes.
 - Transitive: **No.** We have $|0 - 0.6| < 1$ and $|0.6 - 1.2| < 1$, but $|0 - 1.2| = 1.2 \not< 1$.

0.4.3 Equivalence Classes

An equivalence relation partitions a set into disjoint groups of “equivalent” elements. Each such group is called an **equivalence class**.

Definition 0.4.3 — Equivalence Class

Let $\sim$ be an equivalence relation on A . The **equivalence class** of an element $a \in A$ is the set of all elements equivalent to a :

$$[a] = \{b \in A : b \sim a\}$$

The element a is called a **representative** of the class $[a]$.

Example. (Examples of Equivalence Classes)

We illustrate equivalence classes with two examples:

1. **Congruence modulo 3:** The equivalence classes of $\mathbb{Z}$ under $\equiv \pmod{3}$ are:

$$[0] = \{\dots, -6, -3, 0, 3, 6, 9, \dots\}$$

$$[1] = \{\dots, -5, -2, 1, 4, 7, 10, \dots\}$$

$$[2] = \{\dots, -4, -1, 2, 5, 8, 11, \dots\}$$

Note that $[0] = [3] = [6] = \dots$ and $[1] = [4] = [7] = \dots$. Different representatives can name the same class.

2. **Same absolute value on $\mathbb{R}$:** The equivalence class of 2 is $[2] = \{-2, 2\}$. The equivalence class of 0 is $[0] = \{0\}$.

Theorem 0.4.1 — Equivalence Classes Partition the Set

Let $\sim$ be an equivalence relation on A . Then:

1. Every element belongs to some equivalence class: $a \in [a]$.
2. Two equivalence classes are either equal or disjoint: $[a] = [b]$ or $[a] \cap [b] = \emptyset$.
3. The union of all equivalence classes is A .

In other words, the equivalence classes form a **partition** of A .

Proof.

- (1) By reflexivity, $a \sim a$, so $a \in [a]$.
- (2) Suppose $[a] \cap [b] \neq \emptyset$, and let $c \in [a] \cap [b]$. Then $c \sim a$ and $c \sim b$. By symmetry, $a \sim c$. By transitivity, $a \sim b$.

Now we show $[a] = [b]$. Let $x \in [a]$, so $x \sim a$. Since $a \sim b$, by transitivity $x \sim b$, so $x \in [b]$. Thus $[a] \subseteq [b]$. By symmetry, $[b] \subseteq [a]$, so $[a] = [b]$.

- (3) Follows from (1): every element a is in $[a]$.

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Definition 0.4.4 — Quotient Set

Let $\sim$ be an equivalence relation on A . The **quotient set** (or **quotient of A by $\sim$**) is the set of all equivalence classes:

$$A/\sim = \{[a] : a \in A\}$$

Example. (Quotient Sets)

The following are examples of quotient sets:

1. $\mathbb{Z}/\equiv_3 = \{[0], [1], [2]\}$ has three elements. This is often written $\mathbb{Z}/3\mathbb{Z}$ or $\mathbb{Z}_3$.

2. The quotient of $\mathbb{R}$ by “same absolute value” is in bijection with $[0, \infty)$ —each equivalence class is determined by its absolute value.

0.4.4 Equivalence Relations in Linear Algebra

The following equivalence relations will appear throughout your study of linear algebra:

Example. (Matrix Similarity)

On $M_n(F)$, define $\mathbf{A} \sim \mathbf{B}$ if there exists an invertible matrix $\mathbf{P} \in M_n(F)$ such that $\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$. This is called **similarity**.

- Reflexive: $\mathbf{A} = \mathbf{I}^{-1}\mathbf{A}\mathbf{I}$, so $\mathbf{A} \sim \mathbf{A}$.
- Symmetric: If $\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$, then $\mathbf{A} = \mathbf{P}\mathbf{B}\mathbf{P}^{-1} = (\mathbf{P}^{-1})^{-1}\mathbf{B}(\mathbf{P}^{-1})$, so $\mathbf{B} \sim \mathbf{A}$.
- Transitive: If $\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ and $\mathbf{C} = \mathbf{Q}^{-1}\mathbf{B}\mathbf{Q}$, then $\mathbf{C} = \mathbf{Q}^{-1}\mathbf{P}^{-1}\mathbf{A}\mathbf{P}\mathbf{Q} = (\mathbf{P}\mathbf{Q})^{-1}\mathbf{A}(\mathbf{P}\mathbf{Q})$.

Similar matrices represent the same linear transformation in different bases. They share many properties: determinant, trace, eigenvalues, and characteristic polynomial.

Example. (Row Equivalence)

Two matrices are **row equivalent** if one can be obtained from the other by a sequence of elementary row operations. This is an equivalence relation on $M_{m \times n}(F)$.

Remark. When you encounter a new equivalence relation, ask: “What properties do equivalent objects share?” For similarity, the answer includes eigenvalues and determinant. These shared properties are called **invariants** of the equivalence relation.